

T, channel 7 | geometric fitting audit

198 short-delay trials | 33 per direction | final 500 ms before GO

$m = 3$ | $\tau = 16$ ms for every trial | 468 points per cloud | no decoding

1-torus: planar band

Normalized 3D error: median 0.2448 | IQR 0.1980 to 0.2811

Returned: 198/198 | Converged: 198/198 | Any flag: 91/198

Collapsed-hole flag: 0/198 | Active/near bounds: 58/198

2-torus: donut volume

Normalized 3D error: median 0.0012 | IQR 0.0010 to 0.0016

Returned: 198/198 | Converged: 197/198 | Any flag: 183/198

Collapsed-hole flag: 99/198 | Active/near bounds: 163/198

Both fitters use nonlinear least squares with their unchanged native objectives, bounds, and Huber loss.

The 1-torus routine fits a planar annular band. The 2-torus routine fits a solid donut-shaped tube.

The shared normalization divides squared 3D distances by cloud variance. It does not remove differences in model flexibility.

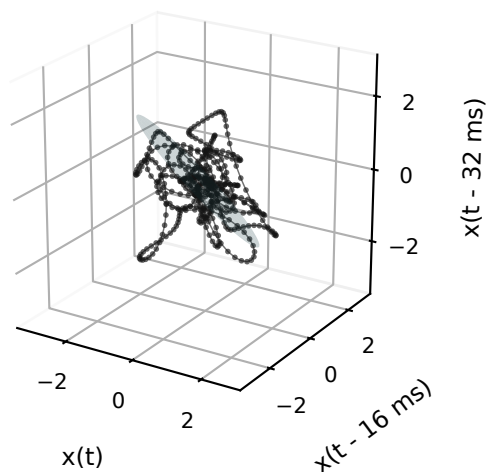
Native coverage and R-squared have different definitions. Numerical convergence or high containment is not proof of torus topology.

The previous 0.80 residual measured a time-parameterized oscillatory trajectory, not these geometric distances.

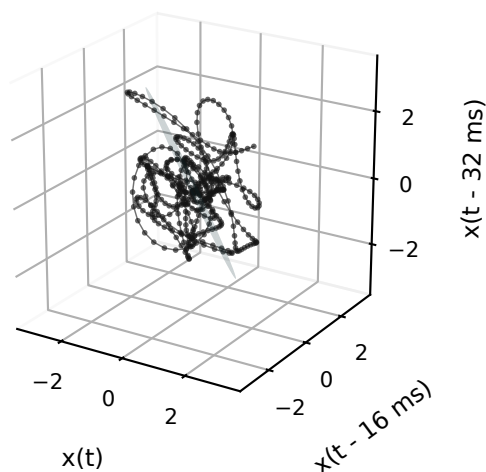
Fixed-delay geometry | matched examples

Lowest original trial ID in each direction; selected without fitting scores

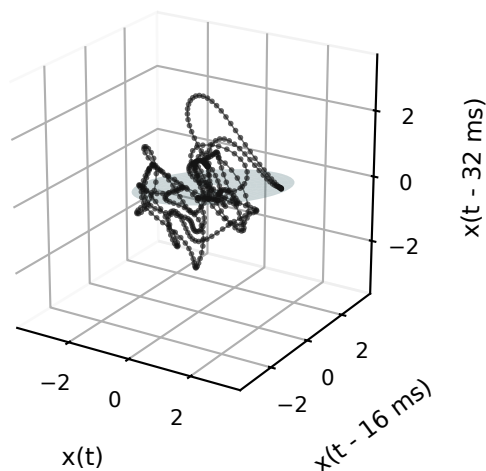
D1 / trial 16 | 1-torus: planar band
at/near bound



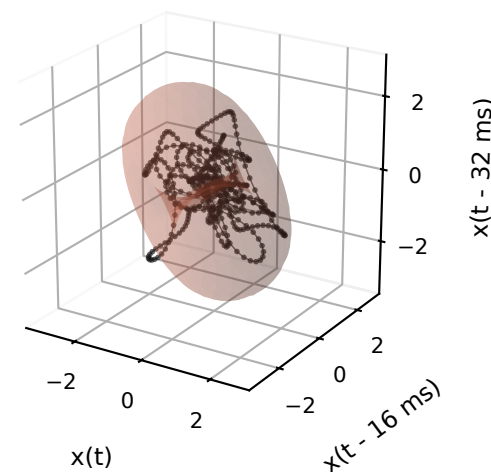
D2 / trial 10 | 1-torus: planar band
no fit flags



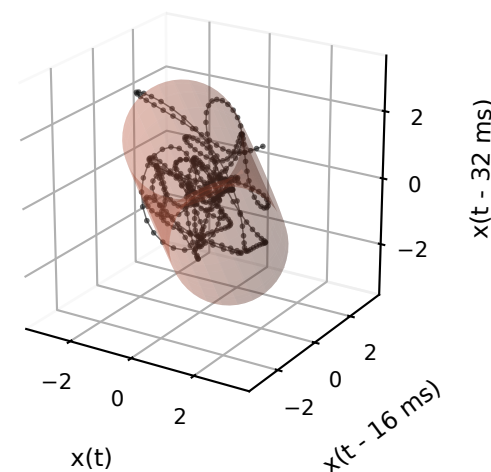
D3 / trial 14 | 1-torus: planar band
no fit flags



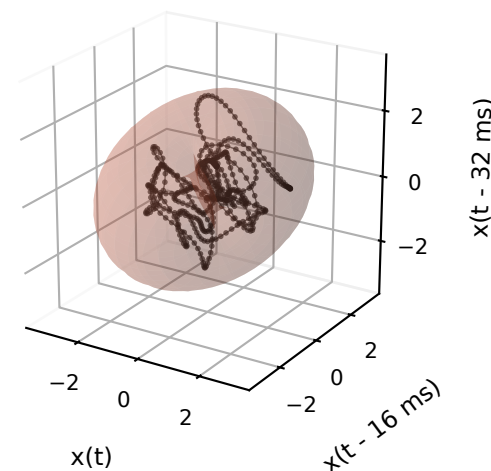
D1 / trial 16 | 2-torus: donut volume
collapsed hole; at/near bound



D2 / trial 10 | 2-torus: donut volume
collapsed hole; at/near bound



D3 / trial 14 | 2-torus: donut volume
collapsed hole; at/near bound

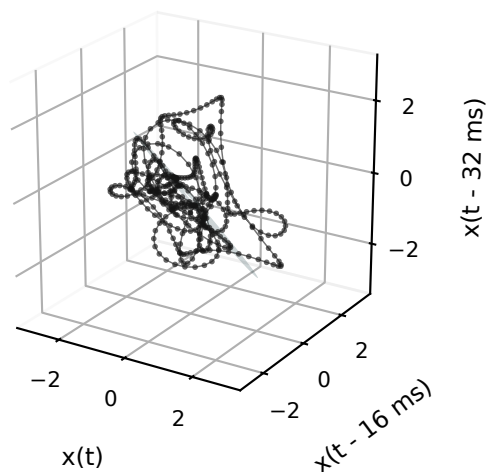


Original lag coordinates; no PCA reduction. Both fits use all 468 points.
Tube overlays are parametric sweeps, not volume boundaries; they can self-intersect. Not a topology test.

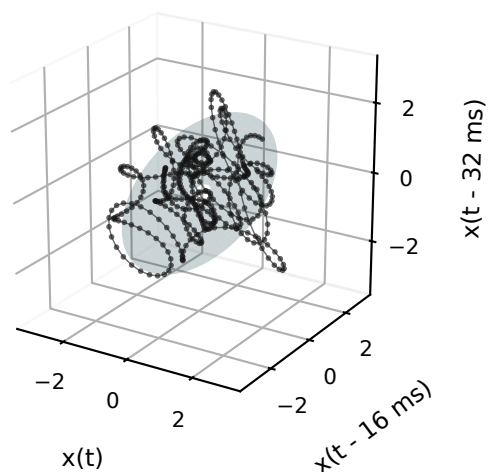
Fixed-delay geometry | matched examples

Lowest original trial ID in each direction; selected without fitting scores

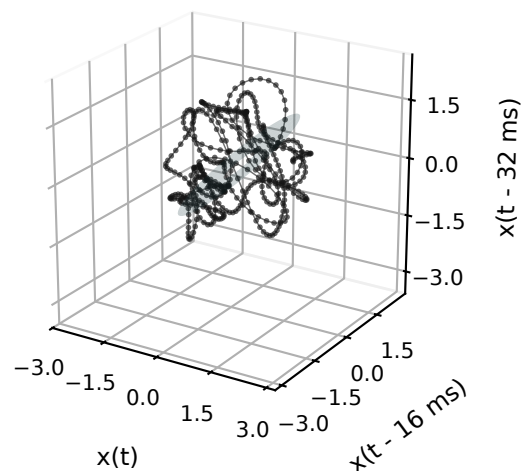
D4 / trial 17 | 1-torus: planar band
no fit flags



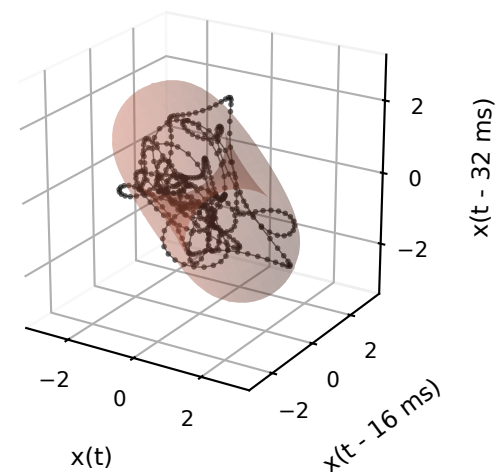
D5 / trial 6 | 1-torus: planar band
no fit flags



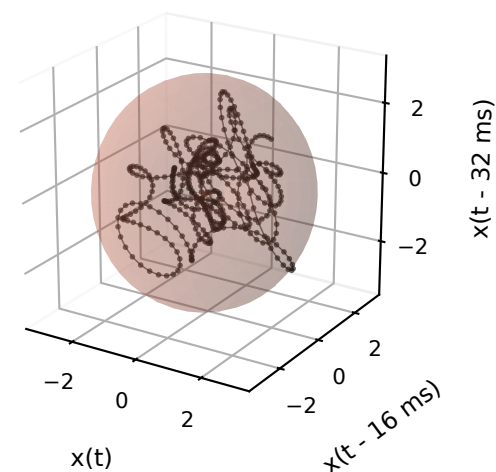
D6 / trial 9 | 1-torus: planar band
no fit flags



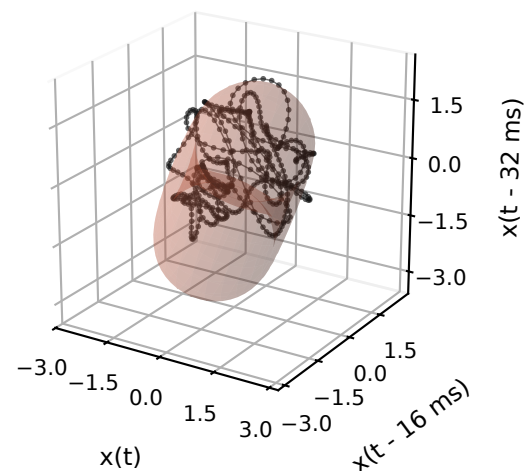
D4 / trial 17 | 2-torus: donut volume
at/near bound



D5 / trial 6 | 2-torus: donut volume
collapsed hole; at/near bound; near-circular axes

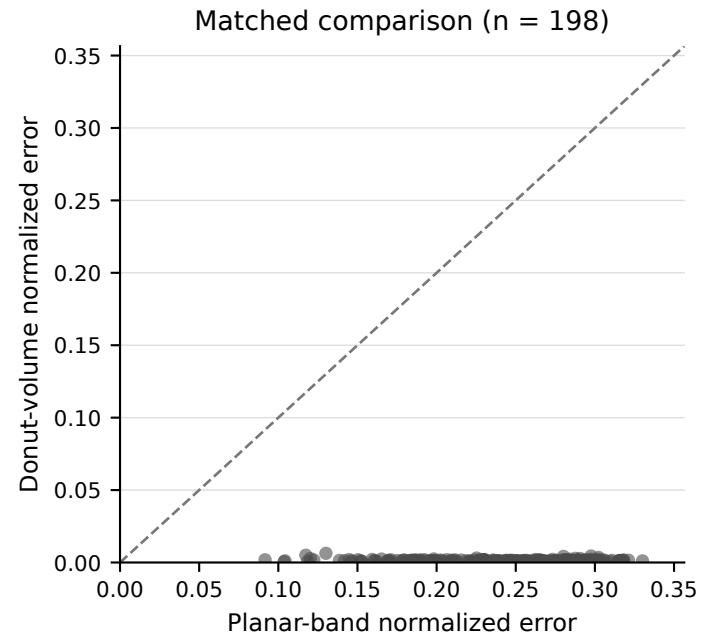
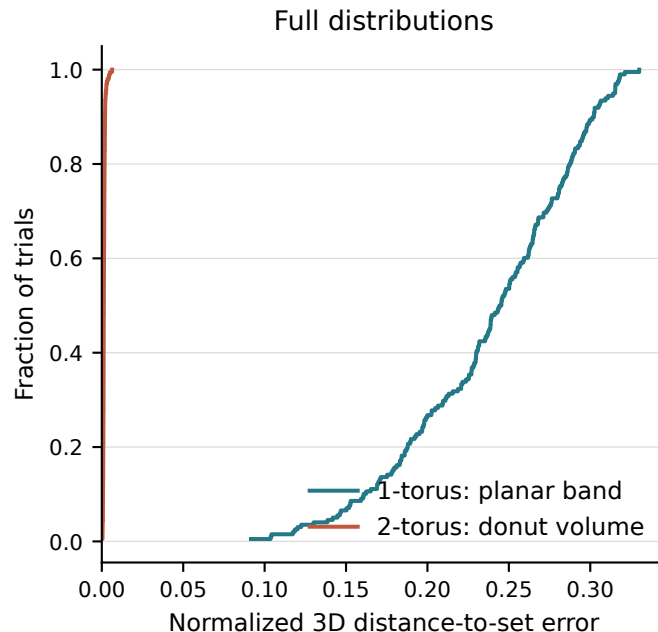
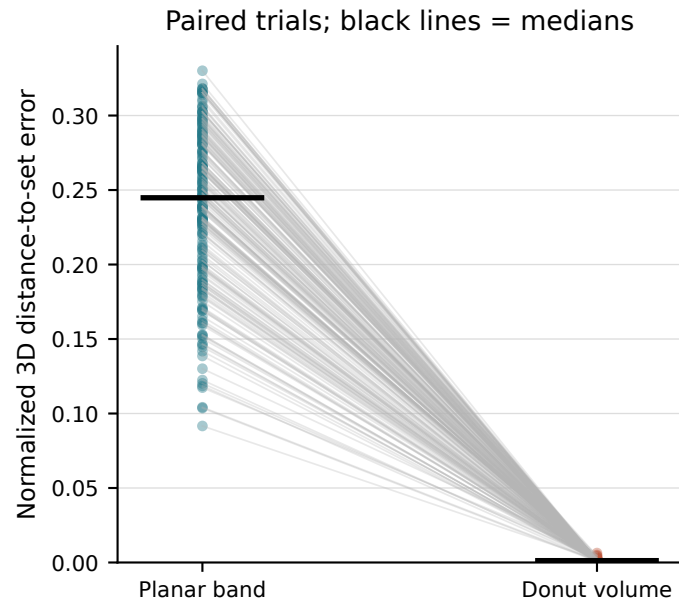


D6 / trial 9 | 2-torus: donut volume
collapsed hole; at/near bound



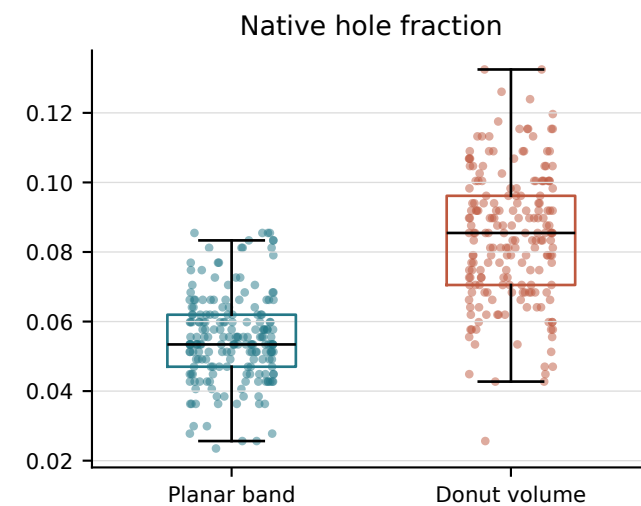
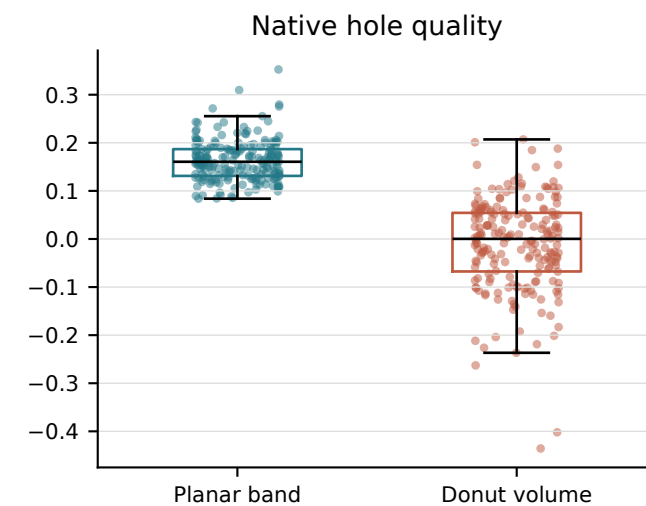
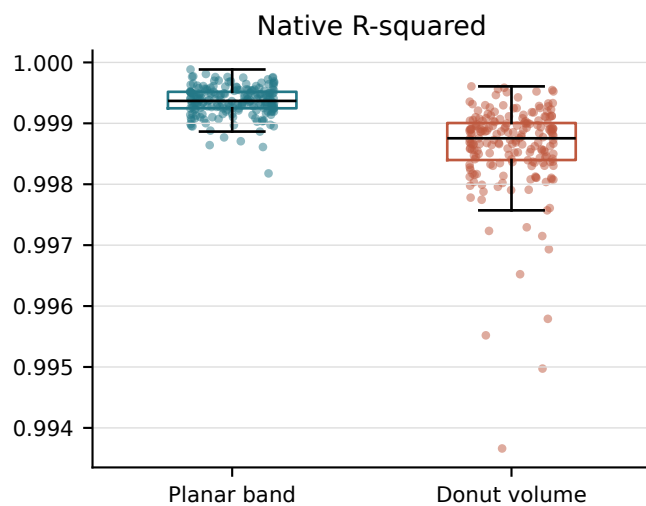
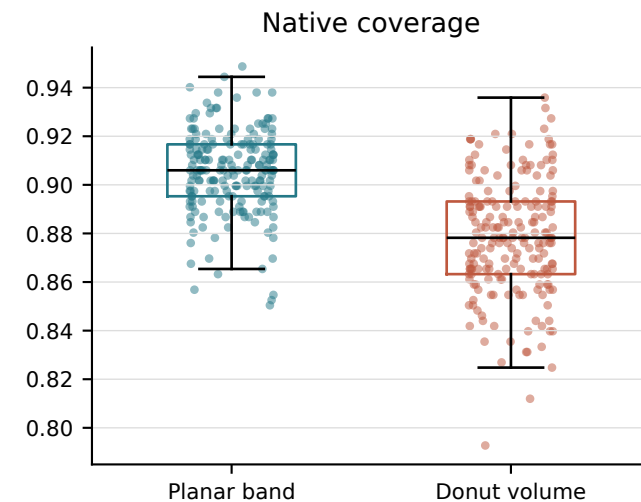
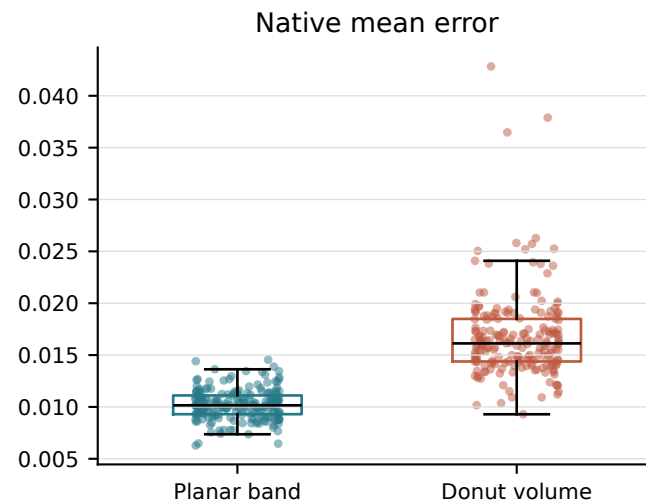
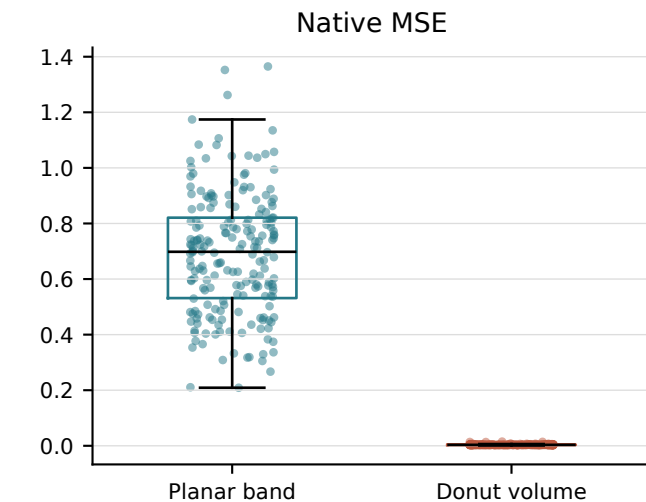
Original lag coordinates; no PCA reduction. Both fits use all 468 points.
Tube overlays are parametric sweeps, not volume boundaries; they can self-intersect. Not a topology test.

Geometric fitting errors | all finite returned fits retained



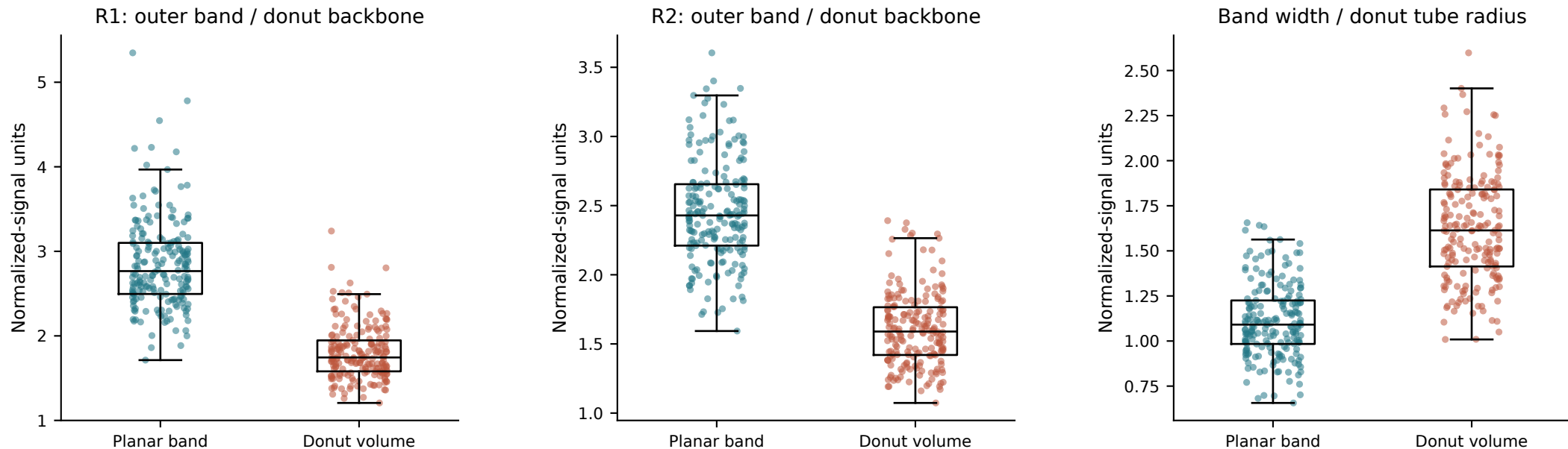
Different fitted sets have different flexibility. Lower containment error does not prove a 2-torus.
These are native approximate geometric distances, not time-trajectory residuals or prediction errors.

Native metrics | definitions differ between fitters



Band: coverage, mean error, and R-squared ignore out-of-plane offsets; MSE includes them.
Donut: errors measure only points outside the solid tube. Neither coverage score establishes topology.

Fitted geometric parameters | native definitions



Band minor_radius = average radial half-width, not out-of-plane thickness.
Complete centers, axes, inner radii, bounds, and optimizer diagnostics are saved per trial.